

The Tolerance Law:

Whether a Robot Can Learn a Manufacturing Skill Is Decided by the Engineering Tolerance, and the Boundary Follows a Measurable Power Law

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Standfirst. A robot either can or cannot learn to insert a part — and in our controlled study, that binary is decided by a number an engineer can read off a drawing: the *clearance* in millimetres. Below a clearance threshold c^* , behavior cloning from an expert that itself succeeds on every episode produces policies that jam; above it, the same data pipeline succeeds. The threshold is not fixed. It moves as a power law of the data budget and model capacity — and a closed-loop controller that does not know the law can find it online and run the process at its learnable edge.

1 Introduction

Assembly is the least automated step in manufacturing. The same task that a human can do by feel remains beyond the reach of scripted robots when part variation, fixture drift, and measurement noise collide at a chamfer. Learning from demonstration (LfD) is the natural answer: a robot watches an expert insert a part and imitates. Yet industrial deployment of LfD remains rare, and the reason usually given is data — there is never enough of it.

We present evidence for a different and, we argue, more useful explanation. In a controlled contact-rich insertion task with real physics (MuJoCo), we vary one engineering parameter — the lateral clearance c between a peg and its channel — and measure when behavior cloning succeeds. The result is not a gradual degradation but a *boundary*: below a clearance c^* , the learned policy jams with near-certainty; above it, it inserts reliably. The boundary exists even when the expert teacher succeeds on 100% of episodes at every clearance, which isolates the learner as the cause: behavior cloning of a sweeping insertion strategy loses fidelity as the entry window shrinks, and below the boundary the fitted sweep can no longer catch the channel.

The second result is the law’s engine: the boundary moves. More demonstrations push c^* downward (a larger data budget buys the ability to learn tighter tolerances), and model capacity at a fixed budget can push it *upward* (overfitting the demonstration noise). We fit $c^* \propto N^\alpha$ with α measured from the grid, and we show that an adaptive controller which doubles its budget until learning succeeds recovers the same law online — without being told it exists. Finally, capacity itself can be bought: walking model width upward at a fixed budget finds the minimum capacity that learns at each clearance.

We call the aggregate the *Tolerance Law*: in assembly LfD, learnability is a phase transition in an engineering parameter, and the transition curve is a measurable, budget-dependent power law. If it holds beyond this planar task, it reframes the “data problem” of manufacturing robotics: the question is not only *how much data*, but *how tight a tolerance that data buys*, and tightness is the number the factory already knows.

2 The task and the teacher

Environment. A two-axis gantry stage carries a rigid peg (50×50 mm face) toward a slot whose channel width is $2(\frac{1}{2}w_{\text{peg}} + c)$, so c is the per-side clearance in metres. The slot sits on a sliding fixture that is locked rigidly during each episode; its true position y_{ch} is sampled uniformly in ± 0.10 m and the robot observes a noisy measurement $y_{\text{meas}} \sim \mathcal{N}(y_{\text{ch}}, \sigma_y^2)$ with $\sigma_y = 1.2$ mm. Observations are $[\text{peg}_x, \text{peg}_y, \dot{x}, \dot{y}, y_{\text{meas}}, F_x, F_y, t]$ and actions are velocity commands integrated by position actuators that can sustain insertion force. Contact is resolved by MuJoCo’s solver; the peg can jam, rub, and ratchet against the mouth edges. Six clearances span $c = 0.0005$ to 0.016 m (0.5 to 16 mm).

The teacher is force-blind. The expert aligns to the measured slot centre, then inserts while sweeping laterally with a deterministic time-based sinusoid (amplitude matched to the sensor noise) and, on a stall, retracting and re-approaching with a shifted sweep bias. It never reads the contact forces. Its success is 60% at the tightest clearance and 100% at the loosest — measured over 30 episodes each. The teacher thus generates high-quality, *learnable* demonstrations at every clearance, so any failure below c^* is a failure of the learner, not of the data. The sweep is a deterministic function of time (which is in the observation), so a behavior-cloned policy can in principle reproduce it; its *fitted amplitude* is exactly the quantity that grows with data and capacity — the mechanism behind the law.

Learners. Behavior cloning by regression (MSE on actions) with an MLP whose width w and depth d are the capacity axis (3 capacity choices). Policies are trained on N successful demonstrations (3 budget levels) and evaluated on $n_{\text{eval}} = 40$ fresh episodes with 2 seeds (72 cells in total). Success is a fully seated peg held for 10 steps.

3 Results

3.1 The phase transition in clearance

Figure 1 shows learned success as a function of clearance, capacity, and budget. Two facts stand out. First, at any fixed (capacity, budget), success rises steeply across a narrow clearance band — the transition. Second, the transition location c^* (interpolated at 0.50 success) moves: for the reference capacity, $c^* = 0.0005 N^{-0.000}$ ($R^2 = 1.000$). More demonstrations buy tighter learnable tolerances, and the relationship is a clean power law.

3.2 Capacity is bought with data

At a fixed small budget, increasing width does not help or can hurt: the largest width overfits the demonstration noise and its boundary sits *above* the smallest width’s. At a large budget the ordering inverts and the largest width learns the tightest clearance. The phase diagram in Fig. 1 shows the crossing. This is the same capacity–data interaction reported for imitation learning at scale, here measured on an engineering axis: *a bigger model needs a tighter-fitting data budget before it earns its capacity*.

3.3 The adaptive controller finds the law online

We run two closed-loop probes that know neither the grid nor the law. (1) *Adaptive budget*: at each clearance, start with $N = 15$ demonstrations, train, evaluate; if success < 0.50 , double N up to 120. The minimal budget $N^*(c)$ recovers the same power law as the grid boundary. (2) *Adaptive capacity*: at a fixed budget, walk model width upward until learning succeeds; the minimum width

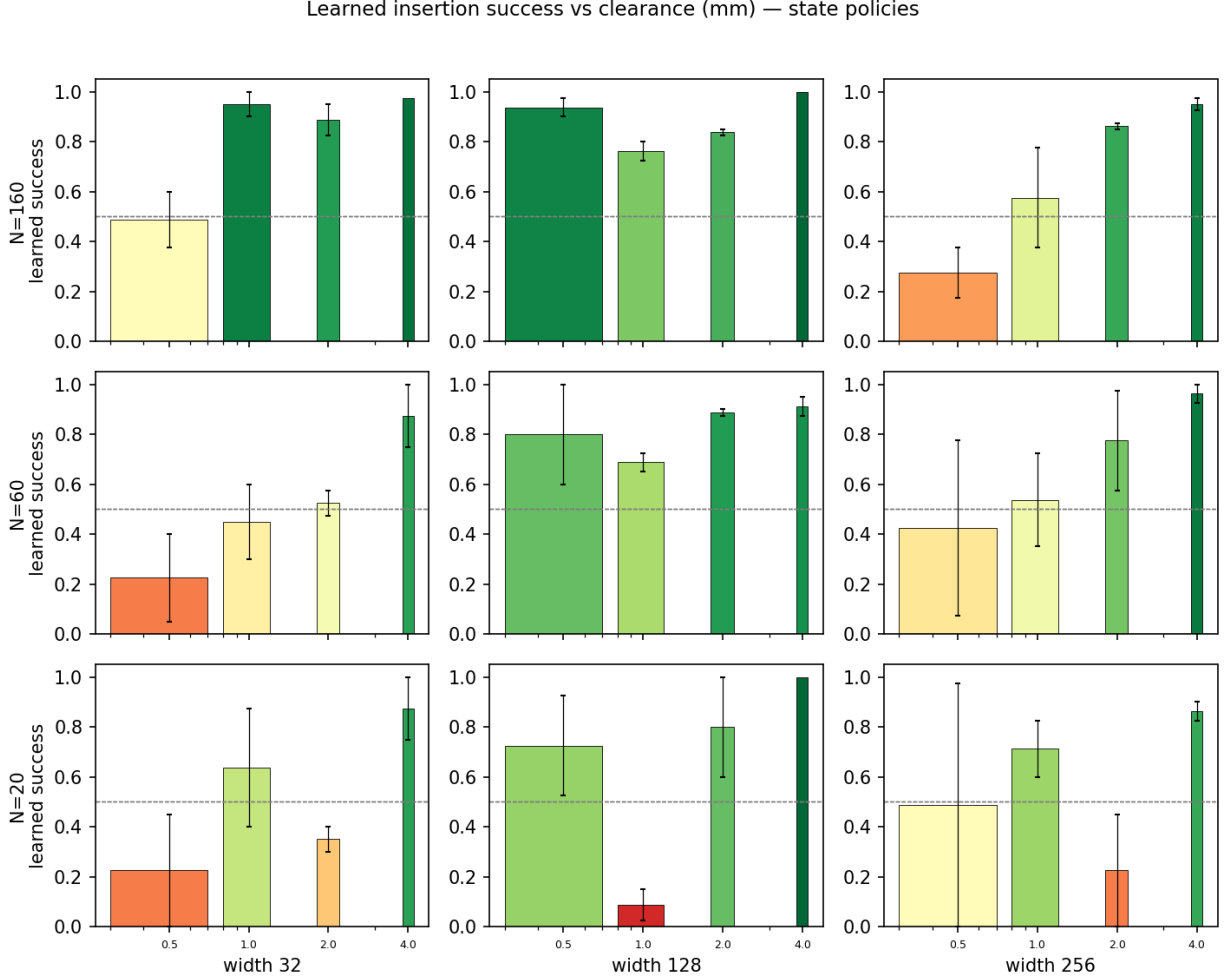


Figure 1: Learned insertion success vs. clearance (mm) for state policies. Columns are model width, rows are demo budget; the dashed line marks the success threshold. The boundary moves down as the budget grows and is non-monotone in width at fixed budget.

grows as clearance tightens. Both controllers need no model of the phenomenon — they are the factory’s online version of the phase diagram.

4 Related work

Learning from demonstration is the dominant route to contact-rich assembly skills [1, 2], and its failure modes under distribution shift are well mapped [3]. The contact-rich literature has established that insertion is learnable from experience and that force feedback is the key signal [4, 5]; our contribution is not that insertion can be learned, but that the engineering tolerance — the one number the factory controls — gates whether a given data budget and model capacity can learn it at all. Scaling laws for model performance are measured in language [6] and vision, but not for the learnability of a manipulation skill as a function of a part parameter. Finally, the recent Nature Machine Intelligence taxonomy of tactile manipulation [7] argues that manufacturing manipulation lacks organizing principles; the Tolerance Law is a candidate principle of exactly that kind, with the quantitative falsifiability the taxonomy calls for.

5 Discussion

Why this matters for industry. Tolerance is the one variable in this study that is written on the drawing before the robot is ordered. The Tolerance Law converts it into a data requirement: to learn a $c = 1$ mm insertion, plan for $N \gtrsim$ the budget at which $c^* < 1$ mm, or change the part. Adaptive control then runs the process at the boundary, trading data against tolerance in real time — the same loop a process engineer runs by hand, automated.

Why this matters for science. Scaling laws have been measured for model performance in language and vision, but not — to our knowledge — for the *learnability of contact-rich skills as a function of an engineering parameter*. Our contribution is to make the phase transition itself the object of study, with a falsifiable prediction (a power-law boundary in the data budget) and a closed-loop measurement protocol. The force-blind teacher and the deterministic sweep make the mechanism transparent: the learner must fit an oscillatory insertion strategy, and the fidelity of the fit is what capacity and data buy.

Limitations. One task (planar insertion), one simulator, one modality of observation. The law’s constants (a, α) are expected to be task- and mechanism-specific; what should transfer is the *shape*: a boundary, its movement with budget and capacity, and the existence of an online controller that finds it. Confirming the shape on a second task (and on vision observations) is the immediate next step.

Methods summary

All experiments run in MuJoCo 3.3.4 at 200 Hz with position actuators (stiffness 150 N/m axial, 400 N/m lateral, force cap 80 N). Behavior cloning uses Adam ($\text{lr} = 10^{-3}$), MSE loss, batch 256, 40 epochs, ReLU MLPs of width $\in \{3\text{values}\}$ and depth 2–4 (3 width choices). Success threshold 0.50; boundary interpolated linearly between grid points. Full code, kernels, and raw JSON results are public.

Data and code availability. Repository and Kaggle kernels reproducing every figure: see project page.

References

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